

19. What is the first step in amino acid catabolism?

- a) Removal of the carbon skeleton
- b) Removal of the nitrogen group
- c) Conversion to glucose
- d) Formation of ketone bodies

Answer: b

20. Which method of nitrogen removal from amino acids involves the formation of glutamic acid without producing free ammonia?

- a) Oxidative deamination
- b) Transamination
- c) Transdeamination
- d) Hydrolysis

Answer: b

21. What is the main mechanism for deamination of most amino acids?

- a) Transamination
- b) Transdeamination
- c) Oxidative deamination
- d) Decarboxylation

Answer: b

22. Which enzyme catalyzes the oxidative deamination of glutamic acid?

- a) Glutamine synthetase
- b) Carbamoyl phosphate synthetase-I
- c) Glutamate dehydrogenase
- d) Transaminase

Answer: c

23. Which amino acids are exclusively ketogenic?

- a) Leucine and Lysine
- b) Tyrosine and Phenylalanine
- c) Tryptophan and Isoleucine
- d) Valine and Methionine

Answer: a

24. What is the fate of glucogenic amino acids?

- a) Conversion to glucose
- b) Formation of ketone bodies
- c) Conversion to acetyl CoA
- d) Formation of neurotransmitters

Answer: a

25. What is the major organ responsible for urea formation?

- a) Kidney
- b) Liver
- c) Brain
- d) Pancreas

Answer: b

26. What is the key enzyme that catalyzes the first step of the urea cycle?

- a) Arginase
- b) Glutaminase
- c) Glutamate dehydrogenase
- d) Carbamoyl phosphate synthetase-I

Answer: d

27. What does the urea cycle utilize from the TCA cycle?

- a) ATP
- b) CO₂
- c) Acetyl CoA
- d) FADH₂

Answer: b

28. What is the normal fasting plasma urea range in humans?

- a) 10–30 mg/dL
- b) 15–40 mg/dL
- c) 20–50 mg/dL
- d) 25–60 mg/dL

Answer: b

29. What happens to plasma urea levels in renal failure?

- a) Decreases
- b) Increases
- c) Remains constant
- d) Fluctuates randomly

Answer: b

30. What is the fate of ammonia in the brain that leads to neurotoxicity?

- a) Conversion to urea
- b) Conversion to glutamine
- c) Excretion in urine
- d) Formation of acetyl CoA

Answer: b

31. How does increased ammonia in the brain affect energy production?

- a) Increases ATP synthesis
- b) Enhances the TCA cycle
- c) Depletes α -ketoglutarate
- d) Promotes oxidative phosphorylation

Answer: c

32. Which neurotransmitter is depleted in the brain due to ammonia intoxication?

- a) Serotonin
- b) GABA
- c) Dopamine
- d) Acetylcholine

Answer: b

33. Which enzyme converts glutamate to glutamine in the brain during ammonia intoxication?

- a) Glutamine synthetase
- b) Glutaminase
- c) Transaminase
- d) Glutamate dehydrogenase

Answer: a

34. Where does the urea cycle begin in the cell?

- a) Cytosol
- b) Mitochondria
- c) Endoplasmic reticulum
- d) Golgi apparatus

Answer: b